Solve volume problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume? Copy or complete the first step.

2. A solid has 5 layers of 24 unit cubes. What is its volume?

3. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume? Copy or complete the first step.

   Answer 240 cubes.

2. A solid has 5 layers of 24 unit cubes. What is its volume?

   Answer 120 cubic units.

3. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

   Answer 420 cm3.

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.

   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

Solve volume problems

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ___________________

1. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

2. A solid has 5 layers of 24 unit cubes. What is its volume?

3. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

4. Solve this independently and show every stage: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

5. Use a second method or representation for: A solid has 5 layers of 24 unit cubes. What is its volume?

6. Check this result using an inverse, estimate or known fact: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

7. Prove or explain your reasoning: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

Teacher answers and checking prompts

1. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer 240 cubes.

2. A solid has 5 layers of 24 unit cubes. What is its volume?
   Answer 120 cubic units.

3. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
   Answer 420 cm3.

4. Solve this independently and show every stage: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer 240 cubes.

5. Use a second method or representation for: A solid has 5 layers of 24 unit cubes. What is its volume?
   Answer Expected result: 120 cubic units. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
   Answer Expected result: 420 cm3. A valid check must be shown.

7. Prove or explain your reasoning: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer 6 cm. The explanation must show why the method works.

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve volume problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ___________________

1. Prove or explain your reasoning: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

2. Prove the answer to this question, rather than only calculating it: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

3. Solve this in two different ways and compare them: A solid has 5 layers of 24 unit cubes. What is its volume?

4. Work backwards from the answer to reconstruct the question: 420 cm3.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.

6. Create a new example that tests the same idea as: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer 6 cm. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer Expected result: 240 cubes. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A solid has 5 layers of 24 unit cubes. What is its volume?
   Answer Expected result: 120 cubic units. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 420 cm3.
   Answer One valid reconstruction is: Find the volume of a cuboid 12 cm by 7 cm by 5 cm. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

6. Create a new example that tests the same idea as: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.